

Synergistic Control Strategy for Enhanced Anti-Jackknifing Stability in Distributed-Drive Articulated Trucks

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Abstract—Electro-hydraulic compound steering systems (EHCSSs) adopted by distributed-drive articulated trucks (DATs) provide the probability of further realizing active steering safety (ASS). However, due to the DATs' articulated mechanical structure characteristics, a singular control strategy frequently cannot effectively solve lateral stability problems such as jackknifing. Hence, this article proposes a novel control strategy integrating active steering, torque vectoring, and differential braking to suppress the jackknife with improved lateral stability and less longitudinal speed loss. First, a nonhierarchical dynamic model fused with EHCSS is established, and the Artemisinin optimization algorithm (AOA) is then applied to model parameter majorization. Afterward, an advanced strategy is developed to suppress the jackknife of DATs. Specifically, the proposed anti-jackknifing control strategy is realized through the active steering of the front tractor wheel to track the ideal trajectory, the torque vectoring control of the rear tractor wheel, and the differential braking of the trailer. Finally, hardware-in-the-loop (HIL) experiments are performed to prove the validity and feasibility. The experiment results illustrate that the proposed strategy is markedly effective in anti-jackknifing. Compared to the control groups, the maximum slip angles of the tractor and trailer are improved by 47.86% and 84.26%, respectively, and the maximum yaw rates are improved by 70.20% and 51.71%.

Index Terms—Active steering, anti-jackknifing control strategy, Artemisinin optimization algorithm (AOA), differential braking, distributed-drive articulated trucks (DATs), torque vectoring.

I. INTRODUCTION

ELECTRIC and automated technologies are driving transformative change in the heavy truck industry, advancing transportation toward greater intelligence and efficiency [1], [2]. Distributed-drive articulated trucks (DATs), powered by electric propulsion, provide substantial advantages over traditional vehicles, including greater maneuverability, enhanced

stability, improved energy efficiency, and superior performance in challenging conditions [3]. However, due to the considerably higher damage resulting from accidents involving DATs compared to passenger cars, there are growing safety concerns associated with the operation of DATs [4], [5]. The articulated mechanical structure of DATs makes them particularly susceptible to jackknifing, which can greatly affect their lateral stability while driving [6]. Jackknifing is a hazardous event in which the trailer of an articulated vehicle swings sharply toward the tractor, resembling the motion of a folding knife [7]. Path tracking control provides effective solutions to mitigate these issues and ensure the safe operation of DATs, a critical topic in both industry and academia [8], [9], [10]. Automotive original equipment manufacturers (OEMs) have developed more efficient control subsystems for path-tracking, including active front steering systems [11], active rear steering systems [12], and torque vectoring control systems [13]. Several scholars have implemented the electro-hydraulic compound steering system (EHCSS) for active steering control [14], [15]. The EHCSS combines the advantages of electric and hydraulic steering systems, offering active steering capabilities for DATs [16]. Deploying the EHCSS is essential for ensuring effective lateral stability control of DATs in challenging operational environments and precise trajectory tracking. However, integrating multiple control strategies with the EHCSS to address jackknifing and stability challenges in DATs under varying operational conditions remains largely unexplored.

The combined control strategy of differential braking and active steering has been demonstrated to enhance the lateral stability of DATs [17]. The controller in [18] employs model predictive control (MPC), using differential braking to conservatively generate additional yaw moments while employing active steering for precise trajectory tracking. By integrating the active steering and differential braking systems, [19] enhances the directional stability of vehicles that turn at high speeds on various road conditions. Additionally, [20] establishes a reconfigurable controller, which integrates multiple strategies of active steering, torque vector, and differential braking. Choi et al. [21] develop a coordinated control of motor-driven power steering torque overlay and differential braking under extreme conditions. In [22], an improved torque vector control and differential braking strategy are proposed. To increase the driving speed and enhance the lateral vehicle stability during cornering, Her et al. [23] propose an integrated

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Designing an effective control strategy for the lateral stability of DATs requires coordinating the interactions between steering, torque vector control, and differential braking while quickly and accurately tracking the ideal trajectory [20], [24]. The structural complexity of the EHCSS presents challenges for coordinated control [14], [25]. A layered time-delay robust control strategy is proposed in [26] to ensure the safety of SbW vehicles. However, hierarchical control design may cause information delays and inaccurate coordination between multiple control strategies, leading to inefficiency [27]. Tian et al. [28] establish a nonhierarchical model, designing a preset performance controller that considers the sideslip angle and yaw rate. Based on a non-hierarchical sensing control structure, a decentralized control scheme is designed in [29] for motion and formation control tasks. Wang et al. [30] employ an adaptive genetic algorithm to optimize the model, enhancing its responsiveness. Therefore, to achieve precise trajectory tracking and minimize system response time while integrating multiple strategies, it is essential to establish a non-hierarchical model that incorporates both the vehicle and steering systems [31].

- 1) A novel anti-jackknifing control strategy is proposed through the active steering of the tractor front wheel, the direct yaw moment control of the tractor rear wheel, and the differential braking of the trailer. Compared to single control strategies, the proposed control strategy achieves more accurate front wheel angle tracking, smaller sideslip angle, and yaw rate with less longitudinal speed loss.
- 2) A novel model fusing a 3 DoF model and an EHCSS model is developed, integrating the Artemisinin optimization algorithm (AOA) for optimizing tire cornering stiffness and an improved fitness function for accurate parameter identification, effectively addressing the limitations of rigid hierarchical structures, allowing for faster system response and improved stability control.

II. VEHICLE DYNAMIC MODEL

Fig. 1 describes the simplification process based on lateral dynamics for DATs. The sideslip angle and the yaw rate of the tractor and trailer are selected as the state variables.

$$I_{1zz}\ddot{\psi}_{1h} = a_{fh}F_{1fh}\cos(\delta_{fh}) - b_{fh}F_{2fh} - c_{fh}F_{4h} \quad (1)$$

$$m_1u_{1h}(\dot{\beta}_{1h} + \dot{\psi}_{1h}) = F_{1fh}\cos(\delta_{fh}) + F_{2fh} + F_{4h} \quad (2)$$

According to the motion of the trailer, the equations are established as follows:

$$I_{2zz}\ddot{\psi}_{2h} = -e_{rh}F_{4h}\cos\Gamma_h - d_{rh}F_{3rh}\cos(\delta_{rh}) \quad (3)$$

$$m_2 u_{2h} (\dot{\beta}_{2h} + \dot{\psi}_{2h}) = F_{3rh} \cos(\delta_{rh}) - F_{4h} \cos \Gamma_h \quad (4)$$

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The motion constraint of DATs can be expressed by the following equation:

$$\dot{\beta}_{1h} - \dot{\beta}_{2h} - \frac{c_{fh}}{u_{1h}} \ddot{\psi}_{1h} - \frac{e_{rh}}{u_{2h}} \ddot{\psi}_{2h} + \dot{\psi}_{1h} - \dot{\psi}_{2h} = 0. \quad (5)$$

The tire lateral force can be approximated by the following equation [32]:

$$\begin{cases} F_{1fh} = k_{1fh} \left(\beta_{1h} + \frac{a_{fh} \dot{\psi}_{1h}}{u_{1h}} - \delta_{fh} \right) \\ F_{2fh} = k_{2fh} \left(\beta_{1h} - \frac{b_{fh} \dot{\psi}_{1h}}{u_{1h}} \right) \\ F_{3rh} = k_{3rh} \left(\beta_{2h} - \frac{d_{rh} \dot{\psi}_{2h}}{u_{2h}} + \delta_{rh} \right) \end{cases} \quad (6)$$

where k_{1fh} , k_{2fh} , and k_{3rh} are the lateral stiffness of the front and rear axle of the tractor and the lateral stiffness of the equivalent shaft of the trailer, respectively. β_{1h} and β_{2h} represent the sideslip angle of tractor and trailer.

In most real-world driving scenarios, especially during high-speed stable driving, the relative motion between the trailer and the tractor is minimal. Since this paper mainly studies the motion of DATs under high-speed conditions, it is reasonable to assume that the longitudinal speeds of the trailer and the tractor are equal. This simplification not only aligns with conventional vehicle dynamics modeling methods but also simplifies the mathematical formulation, making the controller design more practical

$$u_{1h} = u_{2h} = u_h \quad (7)$$

where u_h is the longitudinal speed of DATs.

B. EHCSS Model

EHCSS achieves optimal performance by integrating electronic control units with hydraulic actuators. The electronic control units deliver precise command signals, while the hydraulic actuators generate significant mechanical force. In Fig. 2, the structural diagram of the EHCSS is presented.

The following equation includes the physical model of the EHCSS connected to each of these components [14]:

$$\begin{cases} J_{mh}\ddot{\theta}_{mh} + B_{mh}\dot{\theta}_{mh} = \varsigma T_{eps} + \kappa_{sw}(\theta_{sw} - \theta_{mh}) - F_h l_h \\ m_{zh}\ddot{x}_{zh} + B_{zh}\dot{x}_{zh} = F_{zh} - \frac{T_{ch}}{R_w} + F_{hps} \\ J_{ch}\ddot{\theta}_{ch} + B_{ch}\dot{\theta}_{ch} = T_{ch} - T_P \end{cases} \quad (8)$$

where J_{mh} represents the moment of inertia. B_{mh} is the damping coefficient of the steering screw. θ_{mh} and θ_{sw} denote the steering screw angle and the steering wheel angle, respectively. ζ is the reduction gear ratio. T_{eps} represents the output torque of the electric power. κ_{sw} denotes the steering shaft stiffness. F_h and l_h are the axial working load of the steering screw and the axial working load of the steering screw, respectively. m_{zh} is the steering nut quality. x_{zh} denotes the displacement of the steering nut. B_{zh} is the damping coefficient of the steering sector. F_{zh} is the nut axial force. T_{ch} represents the steering sector torque, R_w denotes the sector pitch radius. F_{hps} represents the output power of the hydraulic power model.

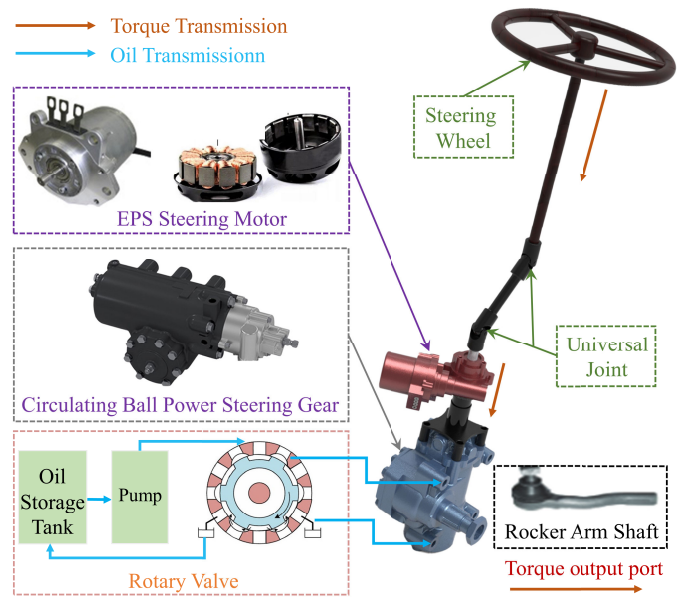


Fig. 2. Structural diagram of the EHCSS.

J_{ch} and B_{ch} are the moment of inertia and the damping coefficients of the steering sector, respectively. T_P is the equivalent moment on the rocker shaft.

The flow and pressure at the rotary valve port are determined by the following equation:

$$\begin{cases} Q_\sigma = C_n A_\sigma \sqrt{(2\Delta P_\sigma / \chi)} \text{sign}(\Delta P_\sigma) \\ Q_A = A_p \frac{dx_r}{dt} \end{cases} \quad (9)$$

$$\begin{cases} P_A - P_B = \frac{\chi}{\text{SC}_q^2} \left(((Q_t - Q_A)/A_1)^2 - ((Q_t + Q_A)/A_2)^2 \right) \\ P_t = \frac{\chi}{\text{SC}_q^2} \left(((Q_t + Q_A)/A_1)^2 + ((Q_t - Q_A)/A_2)^2 \right) \end{cases} \quad (10)$$

$$F_{hps} = A_p (P_A - P_B) \quad (11)$$

where A_σ and Q_σ represent the throttle area and flow of σ port, C_n represents the discharge coefficient, ΔP is the pressure drop, χ represents the density of oil, A_p is the effective area of the piston of the hydraulic cylinder, Q_A denotes the flow rate of power cylinder, P_A and P_B are the power cylinder inlet and outlet pressure.

The transmission ratios in a steering system are determined by three fundamental relationships. First, the interaction between the steering screw and the steering nut is governed by the screw pitch, which converts rotational motion into linear displacement. Second, the relationship between the steering nut and the steering sector is defined by the effective radius, facilitating the transition from linear to rotational movement. Finally, the ratio between the steering sector and the front wheels dictates how rotational motion is translated into wheel

angle, controlled by a specific ratio factor

$$\begin{cases} \frac{\ddot{\theta}_{mh}}{\ddot{x}_{zh}} = \frac{\dot{\theta}_{mh}}{\dot{x}_{zh}} = \frac{\theta_{mh}}{x_{zh}} = \frac{2\pi}{p} \\ \frac{\ddot{x}_{lm}}{\ddot{\theta}_{ch}} = \frac{\dot{x}_{lm}}{\dot{\theta}_{ch}} = \frac{x_{lm}}{\theta_{ch}} = R_w \\ \frac{\ddot{\theta}_{ch}}{\ddot{\delta}_{fh}} = \frac{\dot{\theta}_{ch}}{\dot{\delta}_{fh}} = \frac{\theta_{ch}}{\delta_{fh}} = i_{ch} \end{cases} \quad (12)$$

where p represents the screw pitch.

According to the above equations, (8) can be rewritten as follows:

$$\ddot{\delta}_{fm} = \frac{1}{J_{eh}} \left[\zeta T_{eps} + \kappa_{sw} (i - \kappa \delta_{fm}) + T_{ch} - T_P - \frac{T_{ch}}{R_{rw}} + F_{hps} - F_h l_h - B_{eh} \dot{\delta}_{fm} \right] \quad (13)$$

where $J_{eh} = ((2\pi/P)R_w J_{mh} + m_{zh} R_w + J_{ch})h_{ch}$, $B_{eh} = ((2\pi/P)R_w B_{mh} + B_{zh} R_w + B_{ch})i_{ch}$, $\kappa = (2\pi/P)R_w i_{ch}$.

C. Nonhierarchical Model

An integrated model fused with a 3-DoF model and an EHCSS model has been established here, which can respond faster to system changes and improve system stability. The control quantity is $\mathbf{u} = [T_{eps}]$. The state space equation is shown below

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} + \mathbf{C} \quad (14)$$

where

$$\begin{aligned} \mathbf{x} &= [\beta_{1m} \quad \dot{\psi}_{1m} \quad \beta_{2m} \quad \dot{\psi}_{2m} \quad \delta_{fm} \quad \dot{\delta}_{fm}]^T \\ \mathbf{A} &= \mathbf{M}^{-1}\mathbf{N}, \quad \mathbf{B} = \mathbf{M}^{-1}\mathbf{P}_1, \quad \mathbf{C} = \mathbf{M}^{-1}\mathbf{Q} \\ \mathbf{M} &= \begin{bmatrix} m_1 u_{1h} & 0 & m_2 u_{2h} & 0 & 0 & 0 \\ m_1 u_{1h} c_{fh} & I_{1zz} & 0 & 0 & 0 & 0 \\ 0 & 0 & m_2 u_{2h} e_{rh} & -I_{2zz} & 0 & 0 \\ 1 & -c_{fh}/u_{1h} & -1 & -e_{rh}/u_{2h} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\ \mathbf{N} &= \begin{bmatrix} \mu_{11} & \mu_{12} & k_{3rh} & \mu_{14} & -k_{1fh} & 0 \\ \mu_{21} & \mu_{22} & 0 & 0 & -k_{1fh}(a_{fh} + c_{fh}) & 0 \\ 0 & 0 & \mu_{33} & \mu_{34} & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\kappa_{sw}\kappa/J_{eh} & \mu_{56} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\ \mathbf{P}_1 &= [0 \ 0 \ 0 \ 0 \ 0 \ \zeta/J_{eh}]^T \\ \mathbf{Q} &= [0 \ 0 \ 0 \ 0 \ 0 \ \gamma_{16}]^T \\ \mathbf{P}_2 &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \zeta/J_{eh} \\ 0 & 1/I_{1zz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/I_{2zz} & 0 & 0 & 0 \end{bmatrix}^T \\ \mu_{11} &= k_{1fh} + k_{2fh} \\ \mu_{12} &= (a_{fh} k_{1fh} - b_{fh} k_{2fh}) / u_{1h} - m_1 u_{1h} \\ \mu_{14} &= -k_{3rh} d_{rh} / u_{2h} - m_2 u_{2h} \\ \mu_{21} &= k_{1fh} (a_{fh} + c_{rh}) + k_{2fh} (c_{rh} - b_{fh}) \\ \mu_{22} &= (k_{1fh} a_{fh} (a_{fh} + c_{rh}) + k_{2fh} b_{fh} (b_{fh} - c_{rh})) / u_{1h} \\ &\quad - m_1 u_{1h} c_{fh} \end{aligned}$$

$$\mu_{33} = k_{3rh} (e_{rh} + d_{rh})$$

$$\mu_{34} = -k_{3rh} d_{rh} (e_{rh} + d_{rh}) / u_{2h} - m_2 u_{2h} e_{rh}$$

$$\mu_{56} = -B_{eh} / J_{eh}$$

$$\gamma_{16} = \frac{1}{J_{eh}} \left(\kappa_{sw} i + T_{ch} - T_P - \frac{T_{ch}}{R_w} + F_{hps} - F_h l_h \right).$$

III. PARAMETER OPTIMIZATION

Before designing the control strategy, it is essential to ensure that the dynamic model accurately reflects the operating characteristics of the DATs. Among the parameters affecting this model, the lateral stiffness of the tires plays a critical role in determining side-slip and yaw responses. To solve this issue, an optimization process will be conducted before developing the control strategy.

In the optimization section, the selection of tire cornering stiffness is based on the AOA to ensure that the model parameters accurately match the Trucksim simulation data. The chosen range considers model fitting accuracy, global search and local optimization strategies, and engineering feasibility, ensuring that the optimized parameters enhance system stability while complying with actual vehicle design standards [33]. The algorithm's implementation process is detailed below.

1) *Initialization*: The algorithm identifies the "infection region" within the problem space, representing the initial parameter setting and problem definition stage

$$\mathbf{A}_{M,F} = \mathbf{H} + \mathbf{T}(\mathbf{Q} - \mathbf{H}) = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,F} \\ a_{2,1} & a_{2,2} & \dots & a_{2,F} \\ \vdots & \vdots & \ddots & \vdots \\ a_{M,1} & a_{M,2} & \dots & a_{M,F} \end{bmatrix} \quad (15)$$

where $\mathbf{A}$ is the initial population; $\mathbf{M}$ is the search agent; F is the multidimensional component of the search agent; $\mathbf{Q}$ and $\mathbf{H}$ are the boundaries of the solution space, respectively; R is a sequence of random numbers ranging from 0 to 1.

2) *Comprehensive Elimination*: The strategy for the comprehensive elimination phase is represented by the following equation:

$$\begin{cases} a_{x,y}^{t+1} = a_{x,y}^t + c a_{x,y}^t (-1)^t, & \text{rand} < 0.5 \\ a_{x,y}^{t+1} = a_{x,y}^t + c \cdot \text{best}_y^t (-1)^t, & \text{rand} > 0.5 \end{cases}, \quad r_1 < K \quad (16)$$

where $a_{x,y}^{t+1}$ and $a_{x,y}^t$ are the search agents before and after the update, respectively; best_y^t is the current optimum; r_1 is the random number in the range $[0, 1]$; c is the decay index of the drug concentration in the human body; and K is the probability coefficient. The formula to calculate the two relevant variables is provided below

$$\begin{cases} c = 1 \cdot e^{-\left(4 \frac{f}{\text{Max}f}\right)} \\ K = 1 - \frac{f^{1/6}}{\text{Max}f^{1/6}} \end{cases} \quad (17)$$

where f and $\text{Max}f$ are the current and maximum number of iterations of the algorithm, respectively.

3) *Local Elimination*: The algorithm conducts a refined search in potential “high-risk areas,” narrowing the search scope and focusing resources to explore local regions likely containing issues, ensuring precise target coverage. The particle position is calculated using the following equation:

$$a_x^{t+1} = a_{b3}^t + d \cdot (a_{b1}^t - a_{b2}^t), \quad \text{rand} < \text{Fit}_{\text{norm}}(x) \quad (18)$$

where $\text{Fit}_{\text{norm}}(x)$ and b_1, b_2, b_3 are determined by the following formula:

$$\begin{cases} \text{Fit}_{\text{norm}}(x) = \frac{\text{fit}(x) - \min(\text{fit})}{\max(\text{fit}) - \min(\text{fit})} \\ b_1, b_2, b_3 \sim U(1, N), \quad b_1 \neq b_2 \neq b_3. \end{cases} \quad (19)$$

4) *Final Consolidation*: Considering the reactivation of dormant malaria parasites leading to symptom relapse during treatment, a late consolidation phase strategy was introduced to enhance the algorithm’s ability to avoid local optima

$$\begin{cases} a_{x,y}^{t+1} = a_{x,y}^t, & \text{rand} < 0.05 \\ a_{x,y}^{t+1} = \text{best}_{x,y}, & \text{rand} < 0.2 \end{cases} \quad (20)$$

where $\text{best}_{x,y}$ is the subvector of the current optimal solution in the j th dimension.

Algorithm 1 Artemisinin Optimization Pseudo-Code

```

while ( $f < \text{MaxF}$ )
    Calculate the probability K, exponent c.
    For each agent  $i = 1 : N$ 
        For each dimension  $j = 1 : D$ 
            If  $\text{rand} < K$ 
                Update search agent  $a(i, j)$ .
            End If
            Update search agent  $a(i, j)$ .
            Search agent information crossover.
        End For
    End For
    Calculate the fitness fit.
    Update the population and find the optimal.
     $f = f + N$ 
End While

```

In Algorithm 1, the pseudocode clarifies the process of the AOA. The parameter identification of the DATs model aims to find a set of parameters that best fit the Trucksim simulation data by optimizing three key parameters. The fitness function is defined as the error between the calculated results and the Trucksim data. To ensure the accuracy of parameter identification for the entire fusion model, the ideal front wheel angle and its derivative are incorporated into the formula. Additionally, a difference operation is applied between the ideal front wheel angle and its derivative and the outputs from the EHCSS model at each time step. Finally, the consistency between the model’s prediction results and the reference data is assessed using multiple normalized squared error terms

$$\tau = \sum_{i=1}^z \frac{(\bar{\psi}_{k1} - \bar{\psi}_{s1})^2}{(\max(|\bar{\psi}_{s1}|))^2} + \sum_{i=1}^z \frac{(\bar{\psi}_{k2} - \bar{\psi}_{s2})^2}{(\max(|\bar{\psi}_{s2}|))^2}$$

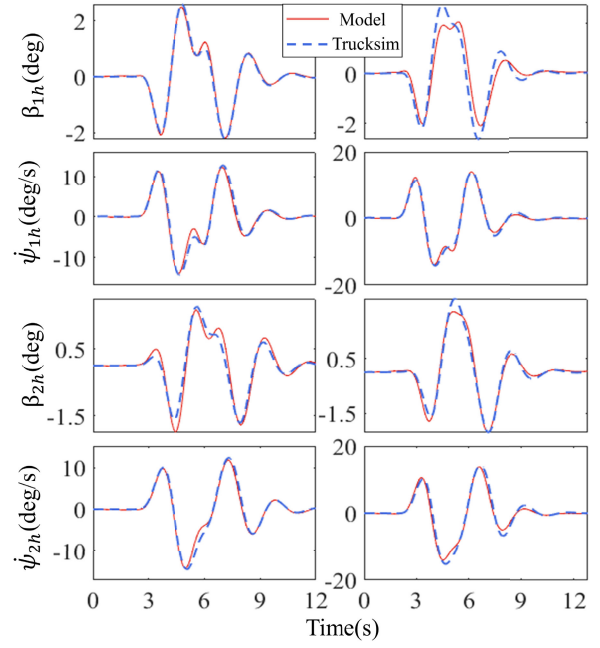


Fig. 3. Identification results (80 and 90 km/h).

$$\begin{aligned} & + \sum_{i=1}^z \frac{(\bar{\beta}_{k1} - \bar{\beta}_{s1})^2}{(\max(|\bar{\beta}_{s1}|))^2} + \sum_{i=1}^z \frac{(\bar{\beta}_{k2} - \bar{\beta}_{s2})^2}{(\max(|\bar{\beta}_{s2}|))^2} \\ & + \sum_{i=1}^z \frac{(\bar{\delta}_{fk} - \bar{\delta}_s)^2}{(\max(|\bar{\delta}_s|))^2} + \sum_{i=1}^z \frac{(\bar{\delta}_{fk} - \bar{\delta}_s)^2}{(\max(|\bar{\delta}_s|))^2} \end{aligned} \quad (21)$$

where $\bar{\psi}_{k1}$, $\bar{\beta}_{k1}$, $\bar{\psi}_{k2}$, and $\bar{\beta}_{k2}$ are the state quantity obtained by model output, and $\bar{\psi}_{s1}$, $\bar{\beta}_{s1}$, $\bar{\psi}_{s2}$, and $\bar{\beta}_{s2}$ are the state quantity obtained from the Trucksim output. $\bar{\delta}_{fk}$ and $\bar{\delta}_s$ are the front wheel angle and the derivative of the front wheel angle output by the model, respectively. $\bar{\delta}_s$ and $\bar{\delta}_s$ are output via Trucksim.

The upper bounds for the corresponding three parameters are set as follows: $[k_{1fh}, k_{2fh}, k_{3rh}]^U = [-le5, -le5, -le5]^U$. The lower bounds are set as follows: $[k_{1fh}, k_{2fh}, k_{3rh}]^L = [-le6, -le6, -le6]^L$.

Double lane change (DLC) at speeds of 80 and 90 km/h, with a road adhesion coefficient of 0.8, are selected for parameter identification. As shown in Fig. 3, a comparison of the parameter identification results with the Trucksim output at speeds of 80 and 90 km/h is presented. The identification results at 80 km/h are $k_{1fh} = -197938.7204$ (N/rad), $k_{2fh} = -574241.1497$ (N/rad), and $k_{3rh} = -249738.3021$ (N/rad). The identification results at 90 km/h are $k_{1fh} = -203480.106$ (N/rad), $k_{2fh} = -556310.5185$ (N/rad), and $k_{3rh} = -238442.5435$ (N/rad).

The overall trends of the sideslip angle and yaw rate are consistent with the Trucksim output, validating the effectiveness of the parameter identification model.

Fig. 4 illustrates the tracking performance of the steering wheel angle at speeds of 80 and 90 km/h.

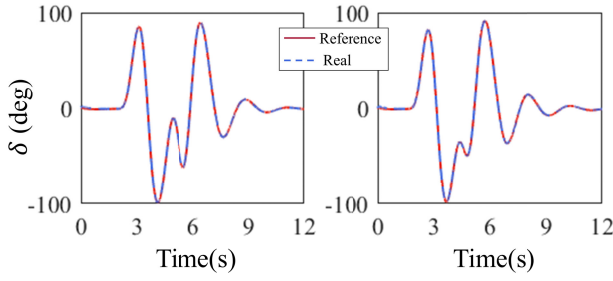


Fig. 4. Comparison of steering wheel (80 and 90 km/h).

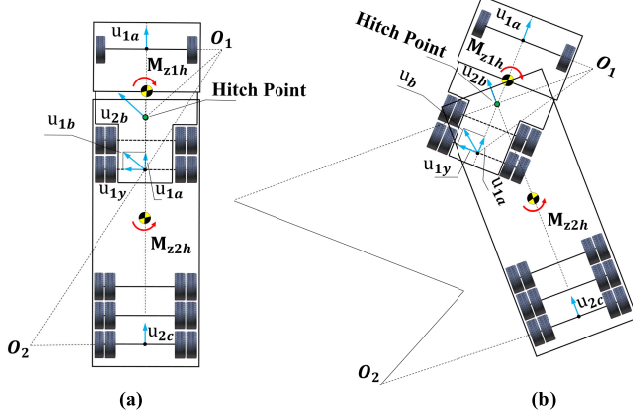


Fig. 5. Principle analysis of jackknifing in DATs. (a) Initial vehicle state. (b) Vehicle jackknife state.

TABLE I

DATs WHEEL TORQUE DISTRIBUTION STRATEGY

Direction	δ_{fm}	e_{ψ}	steering character	tractor	trailer
Left	+	+	under	$+\mathbb{R}_{2,3}, -\mathbb{L}_{2,3}$	$\mathbb{L}_{4,5,6}$
Right	-	+	over	$+\mathbb{R}_{2,3}, -\mathbb{L}_{2,3}$	$\mathbb{L}_{4,5,6}$
Left	0	+	under	$+\mathbb{R}_{2,3}, -\mathbb{L}_{2,3}$	$\mathbb{L}_{4,5,6}$
Right	0	+	over	$+\mathbb{R}_{2,3}, -\mathbb{L}_{2,3}$	$\mathbb{L}_{4,5,6}$
Left	+	+	over	$-\mathbb{R}_{2,3}, +\mathbb{L}_{2,3}$	$\mathbb{R}_{4,5,6}$
Left	0	-	over	$-\mathbb{R}_{2,3}, +\mathbb{L}_{2,3}$	$\mathbb{R}_{4,5,6}$
Left	+	-	over	$-\mathbb{R}_{2,3}, +\mathbb{L}_{2,3}$	$\mathbb{L}_{4,5,6}$
Right	-	-	under	$-\mathbb{R}_{2,3}, +\mathbb{L}_{2,3}$	$\mathbb{R}_{4,5,6}$
Right	0	-	under	$-\mathbb{R}_{2,3}, +\mathbb{L}_{2,3}$	$\mathbb{R}_{4,5,6}$
/	0	0	/	/	/

IV. CONTROL STRATEGY

In this section, a novel anti-jackknifing control strategy is proposed. This strategy utilizes active steering of the front tractor wheels with the EHCSS, torque vectoring of the rear tractor wheels, and differential braking of the trailer.

As shown in Fig. 5(a), when the lateral force exerted on the rear axle of the tractor reaches saturation, the rear axle will slip in the direction of the lateral force. Assuming the lateral force is directed to the left under these circumstances, and the front axle of the tractor remains unaffected by slip, the instantaneous rotation center of the tractor O_1 is determined using the vertical line method. Subsequently, the tractor rotates clockwise around this rotation center, causing the trailer saddle to rotate around the same instantaneous rotation center. The instantaneous steering center of the trailer is O_2 .

TABLE II

KEY PARAMETERS OF DATs AND EHCSS

Symbol	Number (Unit)	Symbol	Number (Unit)
m_{1h}	6595 (kg)	m_{2h}	7570 (kg)
a_{fh}	2.000 (m)	b_{fh}	2.135 (m)
c_{fh}	2.135 (m)	d_{rh}	4.500 (m)
e_{rh}	3.400 (m)	I_{1zz}	34802.6 ($\text{kg} \cdot \text{m}^2$)
I_{2zz}	150000 ($\text{kg} \cdot \text{m}^2$)	P	0.013
R_w	0.042(m)	J_{mh}	0.06 ($\text{kg} \cdot \text{m}^2$)
B_{mh}	0.004($\text{Nm} \cdot \text{s/rad}$)	ς	5
κ_{sw}	204(Nm/rad)	l_h	0.1(m)
m_{zh}	5.44(kg)	B_{zh}	0.031($\text{Nm} \cdot \text{s/rad}$)
J_{ch}	0.031($\text{kg} \cdot \text{m}^2$)	B_{ch}	0.0251($\text{Nm} \cdot \text{s/rad}$)
i_{ch}	0.785		

The insufficient lateral force from the tractor's rear axle causes mutual interaction between the torques generated by the tractor and the trailer. The tractor generates a torque around the hinge point, rotating clockwise, while the trailer generates an opposing torque, rotating counterclockwise. This interaction leads to the "jackknifing" phenomenon of the vehicle, as illustrated in Fig. 5(b). The resulting torque imbalance at the connection point (typically the hinge) misaligns the driving trajectories of the tractor and trailer, potentially leading to severe vehicle instability or loss of control.

A. Controller Design

MPC is fundamentally based on the current state variables and the system model to predict the evolution of state variables over a specified future time horizon. The control strategy involves formulating an objective function, which is typically a cost function, while incorporating system constraints. The optimal control sequence is then determined by solving this optimization problem at each time step. In this section, the MPC controller is designed according to the non-hierarchical model established in Chapter II.

To link the modeling framework with controller design, the MPC state vector and control inputs are directly derived from the nonhierarchical dynamic model. The selected states reflect the yaw and sideslip dynamics and are highly observable and controllable within the EHCSS-integrated DATs system. The control inputs correspond to front-wheel steering, rear electric yaw moment, and trailer differential braking. This direct mapping reduces delay and improves response accuracy. Unlike hierarchical models, the non-hierarchical approach enables unified dynamic prediction, enhancing real-time performance and system stability. Algorithm 2 provides the pseudocode for the MPC-based anti-jackknifing control.

The forward Euler method is used to discretize the state space equation as follows:

$$\mathbf{x}_{\omega}(k+1) = \mathbf{A}_{k\omega}\mathbf{x}_{\omega}(k) + \mathbf{B}_{k\omega}\mathbf{u}_{\omega}(k) \quad (22)$$

where $\mathbf{A}_{k\omega} = \mathbf{I} + \mathbf{A}_2\mathbf{T}_s$, $\mathbf{B}_{k\omega} = \mathbf{B}_2\mathbf{T}_s$.

A discretization linear prediction model is constructed, and the MPC controller predicts the future output by iterating the

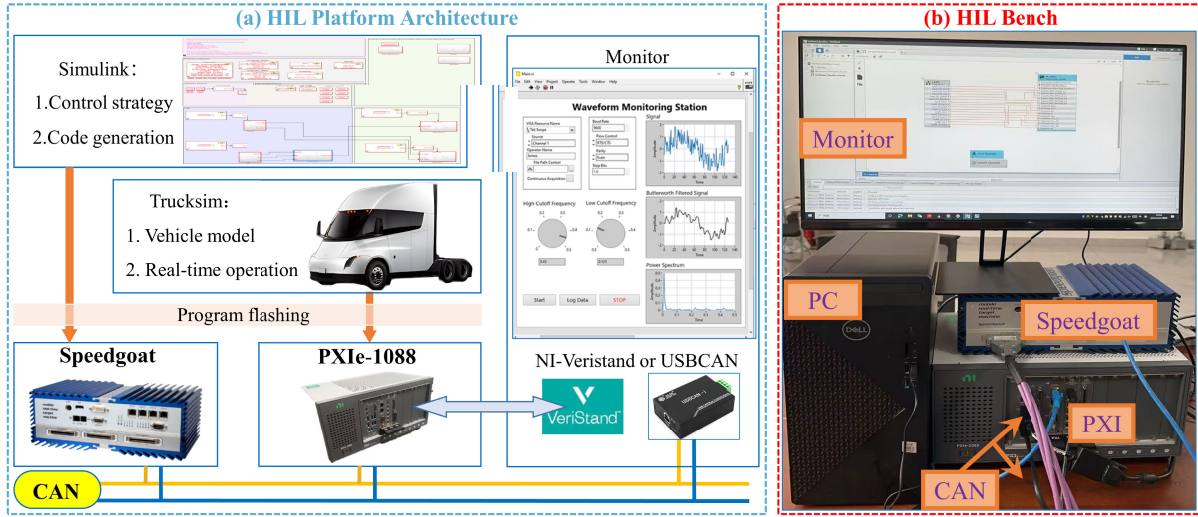


Fig. 6. HIL platform architecture and experimental process.

Algorithm 2 MPC-Based Anti-Jackknifing Control

- 1: Acquire the current state $\mathbf{x}_\omega(k)$ and reference output $\mathbf{y}_d(k)$;
- 2: Solve the QP problem to obtain the optimal input $\mathbf{u}_\omega(k)$;
- 3: Apply $\mathbf{u}_\omega(k)$ to the vehicle system.

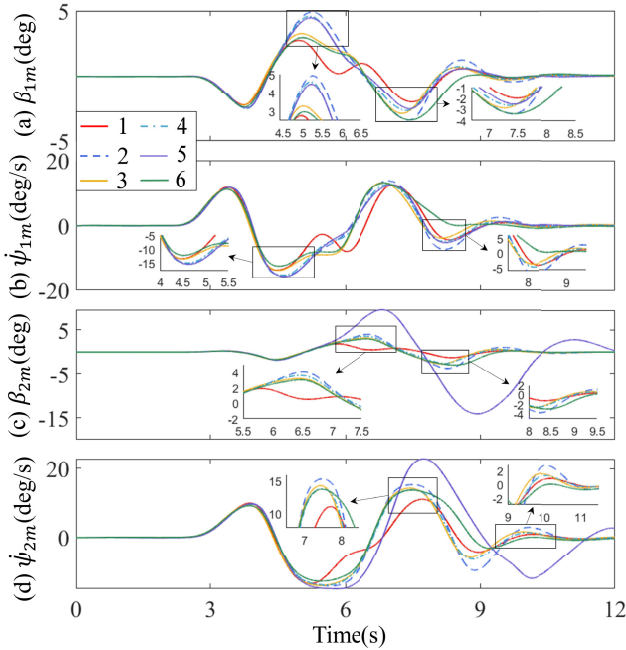


Fig. 7. Comparison of DATs sideslip angle and yaw rate at 80 km/h with a road adhesion of 0.38.

system equations and sorting out the matrix form. N_p and N_c are the prediction time domain and the control time domain, respectively.

$$\mathbf{Y} = \Psi_\mu \mathbf{x}_\omega(k) + \Gamma \mathbf{U}_\mu \quad (23)$$

where

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_\omega(k+l) \\ \vdots \\ \mathbf{y}_\omega(k+N_p) \end{bmatrix}, \quad \mathbf{U}_\mu = \begin{bmatrix} \mathbf{u}_\omega(k) \\ \vdots \\ \mathbf{u}_\omega(k+N_p-1) \end{bmatrix}$$

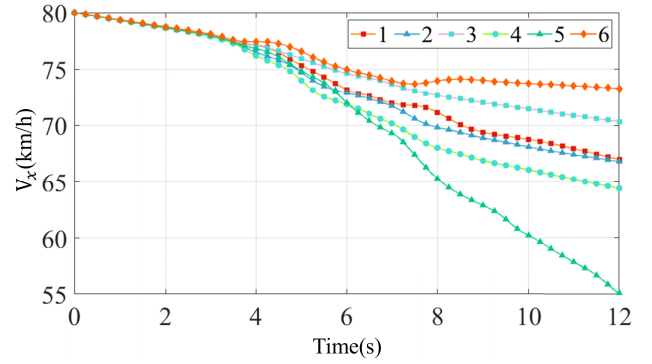


Fig. 8. Speed loss under the evaluated conditions.

$$\Gamma = \begin{bmatrix} \mathbf{C}_\omega \mathbf{B}_{k\omega} & \cdots & 0 \\ \vdots & & \vdots \\ \mathbf{C}_\omega \mathbf{A}_{k\omega}^{N_p-1} \mathbf{B}_{kv} & \cdots & \mathbf{C}_\omega \mathbf{B}_{k\omega} \end{bmatrix}$$

$$\Psi_\mu = \begin{bmatrix} \mathbf{C}_\omega \mathbf{A}_{k\omega} \\ \vdots \\ \mathbf{C}_\omega \mathbf{A}_{k\omega}^{N_p} \end{bmatrix}.$$

The objective function of MPC aims to achieve vehicle path following control as well as yaw stability control while keeping the controller output as small as possible. Considering the system constraints, the performance evaluation function of MPC is designed according to the following formula:

$$J = \sum_{i=1}^{N_p} \left[(\mathbf{y}_e(k+i))^T \mathbf{Q} (\mathbf{y}_e(k+i)) \right] + \sum_{i=1}^{N_c} \left[(\mathbf{u}_\omega(k+i))^T \mathbf{R} (\mathbf{u}_\omega(k+i)) \right] \quad (24)$$

where $\mathbf{y}_e(k+i)$ is the difference between $\mathbf{y}_{d\omega}(k+i)$ and $\mathbf{y}_\omega(k+i)$, $\mathbf{Q}$ and $\mathbf{R}$ are the weight matrices of the output and control quantities, respectively.

To solve the optimal control sequence for MPC, the optimization problem is transformed into a quadratic programming

form. Substituting the system prediction equation into the MPC performance function, we obtain the following equation:

$$J = \|Y_d(k) - \Psi_\mu x_\omega(k) - \Gamma U_\mu\|_{Q_{MPC}}^2 + \|U(k)\|_{R_{MPC}}^2 \quad (25)$$

Make $\epsilon = Y_d(k) - \Psi_\mu x(k)$. The equation can be obtained as follows:

$$\begin{aligned} J &= \|\epsilon - \Gamma U(k)\|_{Q_{MPC}}^2 + \|U(k)\|_{R_{MPC}}^2 \\ &= U^T k H_{mpc} U k + G_{mpc} U k + \epsilon^T Q_{MPC} \epsilon \end{aligned} \quad (26)$$

where H_{mpc} and G_{mpc} are matrices of state and control quantities, respectively. $H_{mpc} = \Gamma^T Q_{MPC} \Gamma + R_{MPC}$, $G_{mpc} = -2\epsilon^T Q_{MPC} \Gamma$.

$\epsilon^T Q_{MPC} \epsilon$ is a constant matrix, and, since this term has no effect on the solution of the optimization problem, it can be ignored. Thus, the MPC optimization problem is transformed into a standard quadratic programming form as shown in the following equation:

$$\begin{aligned} \min J &= \frac{1}{2} U^T(k) H_{mpc} U(k) + G_{mpc} U(k) \\ \text{s.t. } \begin{bmatrix} I \\ -I \end{bmatrix} U(k) &\leq \begin{bmatrix} U_{\max}(k) \\ U_{\min}(k) \end{bmatrix}. \end{aligned} \quad (27)$$

Finally, the calculated control quantity is used to predict the system output of future steps, but in the actual control process, only the first control quantity will be applied

$$u_\omega(k) = [1 \quad 0 \quad \cdots \quad 0] U(k). \quad (28)$$

B. Torque Distribution Strategy

For the tractor, the driving torque distribution strategy is based on the torque plus-minus complementary principle to design the yaw torque distribution controller. The torque plus-minus complementary principle is a control strategy that balances yaw moments by adjusting the torque distribution between the left and right wheels. It works by increasing the torque on one side while reducing it on the other, ensuring that the total driving force remains stable while generating the necessary yaw moment. The equation is shown below

$$\begin{cases} \Delta T_{l2} = \frac{T_d/2 - M_{lad}/l}{2} \\ \Delta T_{r2} = \frac{T_d/2 + M_{lad}/l}{2} \end{cases} \quad (29)$$

where ΔT_{L2} and ΔT_{R2} are the torque distribution values of the left and right wheels of the tractor, respectively, and T_d represents the driving torque of the demand.

For the trailer, the braking torque distribution relationship is realized by the following equation [34]:

$$\begin{cases} \Delta T_{l4} = \frac{2M_{2sum}}{p_3} \frac{F_{zl4}}{F_{zl4} + F_{zl5} + F_{zl6}} r_4 \\ \Delta T_{l5} = \frac{2M_{2sum}}{p_3} \frac{F_{zl5}}{F_{zl4} + F_{zl5} + F_{zl6}} r_5 \\ \Delta T_{l6} = \frac{2M_{2sum}}{p_3} \frac{F_{zl6}}{F_{zl4} + F_{zl5} + F_{zl6}} r_6 \end{cases} \quad (30)$$

where ΔT_{l4} , ΔT_{l5} , and ΔT_{l6} are the torque distribution values of the left of the trailer, p_3 are the wheelbase of the trailer, and F_{zl4} , F_{zl5} , and F_{zl6} are the longitudinal braking force applied sequentially to the left wheel, respectively. Since the left and right equations are formulated in the same form, they will not be explained here.

Due to the implementation complexity of the proposed strategy, it is crucial to consider the interaction between multiple factors simultaneously, which indicates the importance of the wheel torque distribution. The corresponding distribution rules are provided in Table I.

C. Slip Rate Control

The torque vector control can prevent the tractor from wheel locking by generating driving torque, while it is challenging to avert wheel locking for the trailer under the torque distribution strategy due to its numerous wheels. Thus, controlling the slip rate of each trailer wheel is essential to enhance lateral vehicle stability.

The slip rate is calculated using the following equation:

$$\mu_h = (u_h - r_h \omega_h) / u_h \quad (31)$$

where μ_h is the slip rate; u_h is the velocity at the center of the wheel, r_h is the rolling radius of wheels, and ω_h is the wheel angular velocity. The slip rate is regulated with a logical threshold, ensuring it stays within the range of 0.15–0.21.

By giving the threshold value of slip rate, the final wheel braking torque calculation equation is then obtained as follows:

$$\begin{cases} \Delta T_\eta = \Delta T_{\eta-1} + \eta_i t_q, & \mu_h < \mu_{\min} \\ \Delta T_\eta = \Delta T_{\eta-1}, & \mu_{\min} < \mu_h < \mu_{\max} \\ \Delta T_\eta = \Delta T_{\eta-1} - \eta_d t_q, & \mu_h > \mu_{\max} \end{cases} \quad (32)$$

where ΔT_η and $\Delta T_{\eta-1}$ are the magnitude of the braking torque at time η and $(\eta - 1)$, respectively, and t_q is the sampling time. η_i and η_d are identified as the growth rate and decline rate, respectively. The vehicle dynamics parameter information is provided in Table II.

V. EXPERIMENTS AND RESULTS

This section presents the experimental validation of the proposed control strategy. To validate the effectiveness of the control strategy, experimental verification is carried out for various conditions of jackknifing. Five sets of control experiments are conducted for comparison. The results demonstrate that the proposed control strategy outperforms other control strategies under multiple extreme working conditions.

A. Comparison Control Effect of Different Strategy

The architecture of the HIL experimental platform is presented in Fig. 6. The Trucksim vehicle model and road environment are constructed in PXI. To clarify the connection between the HIL experiment, the dynamic model, and the control strategy, the 3-DoF vehicle model and the EHCSS model were integrated into the HIL platform to simulate real vehicle behavior. The proposed control strategy is built in

Speedgoat, the vehicle controller unit. CAN communication connects devices, and the results are monitored by the PC.

The DLC condition is chosen to verify the superiority of the proposed control strategy. Vehicle speeds are set to 80 and 88 km/h, corresponding to road adhesion coefficients of 0.38 and 0.47, respectively. The two selected road conditions represent low-adhesion surfaces that increase the risk of vehicle instability, especially jackknifing. These conditions were chosen to test the control strategy's effectiveness in maintaining lateral stability and preventing vehicle jackknifing under extreme situations. Five groups of control strategies are selected for comparison experiments.

They are, respectively, the following six control strategies.

- 1) Tractor front wheel uses EHCSS to active steering, and the second shaft generates additional yaw moments through torque vector control. The trailer is controlled by differential braking.
- 2) The tractor implements EHCSS for active steering and the trailer for differential braking.
- 3) The tractor uses EHCSS for active steering. The trailer is not controlled.
- 4) Differential braking for tractor and trailer.
- 5) The second axle of the tractor adopts torque vector control, and the trailer implements differential braking.
- 6) The tractor front wheel uses EHCSS to active steering, and the second shaft generates additional yaw moments through torque vector. The trailer is not controlled.

The first comparison is the DAT's sideslip angle and yaw rate under DLC with the vehicle speed of 80 km/h and the road adhesion coefficient of 0.38 in Fig. 7. It shows that the proposed new control strategy outperforms other control strategies. Notably, the method involving tractor torque vector control and trailer differential braking performs the worst, showing larger fluctuations in the slip angle and yaw rate. For the slip angle, the proposed control strategy improves by about 44.14%, and 73.47%, respectively, compared to the worst. For the yaw rate, the proposed control strategy improves by about 70.20% and 51.71%, respectively, compared to the worst.

Longitudinal velocity loss, lateral acceleration, and steering response are essential in comprehensively evaluating the performance and effectiveness of the proposed control strategy. As shown in Fig. 8, the longitudinal speed change process of the six control strategies is given.

From the Fig. 8, it is evident that the longitudinal speed of the proposed strategy is 67.04 km/h at 12 s, with a speed loss rate of 16.20%. The method involving tractor torque vector control and trailer differential braking results in the highest speed loss rate at 31.03%. When the tractor is controlled by active front wheel steering and rear axle torque vector control, the speed at 12 s reaches 73.26 km/h, with the lowest speed loss rate of about 8.43%. Although the proposed control strategy does not have the minimum speed loss rate, it remains within an acceptable range.

As shown in Fig. 9, the articulated angle and lateral acceleration under different control strategies are compared. The proposed new control strategy enables the hinge angle to vary from -5.5 to $+7.3^\circ$, improving about 60.12% compared to

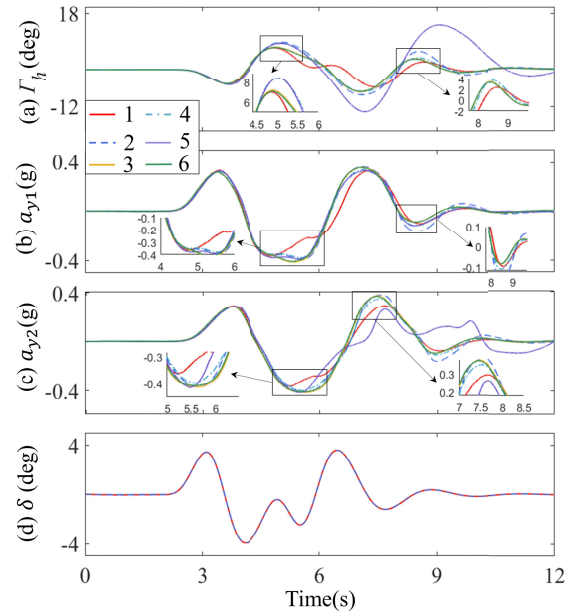


Fig. 9. Comparison of lateral dynamic parameters at 80 km/h with a road adhesion of 0.38.

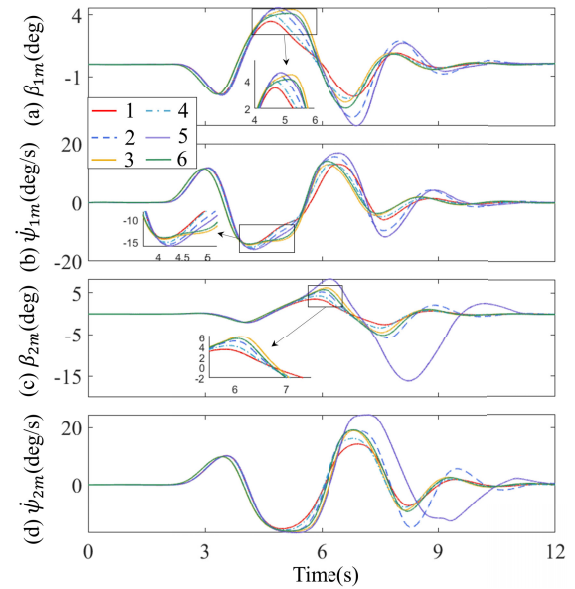


Fig. 10. Comparison of DAT's sideslip angle and yaw rate at 88 km/h with a road adhesion of 0.47.

the worst-performing strategy. The lateral acceleration of the tractor and trailer varies from -0.5 to 0.4 , indicating effective lateral stability with the new control strategy. In terms of trajectory tracking performance, the multiple steering strategies almost perfectly match the ideal front wheel angle trajectory, demonstrating the superior performance of the new control strategy in accurately following the predetermined trajectory. Overall, the newly proposed control strategy significantly improves the stability and accuracy of the system in controlling the hinge angle and lateral acceleration. The comparison results in Fig. 10 verifies the effectiveness of the new strategy.

To verify the rationality of the proposed algorithm, an experiment was conducted at a speed of 88 km/h and a road adhesion coefficient of 0.47. Additionally, the DLC condition is selected for the experiment as it better simulates the vehicle

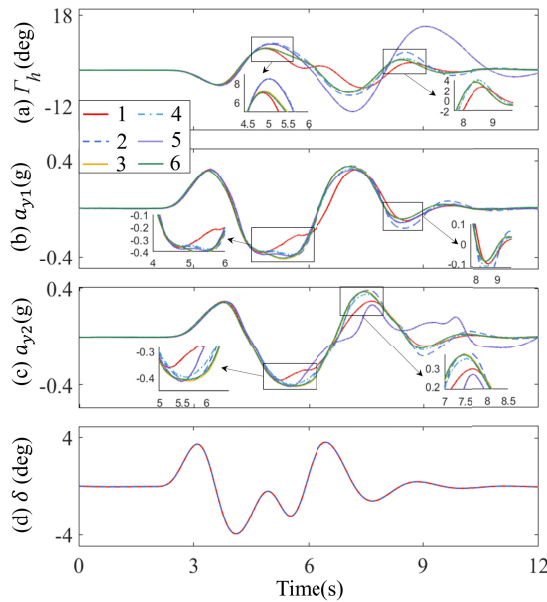


Fig. 11. Comparison of lateral dynamic parameters at 88 km/h with a road adhesion of 0.47.

jackknifing phenomenon. The experiment is conducted under various speeds and adhesion coefficients.

As shown in Fig. 10, comparative experiments with six different strategies were performed. The proposed control strategy consistently outperformed the others. The slip angle of the tractor and trailer was controlled between -3° to 4° and -3° to 4° , respectively. The yaw rate was maintained between -14° to 14° for the tractor and -16° to 16° for the trailer. Compared to the worst strategy, the proposed strategy improves the slip angle of the tractor and trailer by approximately 47.86% and 84.26%, respectively. The yaw rate is improved by 24.27% and 41.57%, respectively.

Here, the articulated angle and lateral acceleration are compared. As shown in Fig. 11, the proposed control strategy maintains the articulated angle between -7° and 8° , with a smaller fluctuation range compared to other strategies. The corresponding lateral acceleration is controlled between -0.43 and 0.4 . Notably, the proposed strategy exhibits greater stability, with fewer fluctuations than other control strategies.

Moreover, the tracking effect of the proposed control strategy is given in Fig. 11. It can be found that the error of the tracking effect of the front wheel angle is the smallest, which conforms to the ideal trajectory given.

VI. CONCLUSION

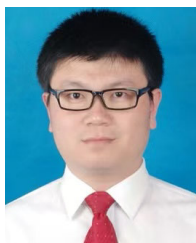
In this study, an integrated control strategy is developed and implemented in the DAT's stability control to suppress jackknifing under extreme conditions. Specifically, the proposed strategy is realized through the active steering of the tractor's front wheel, the torque vectoring of the tractor's rear wheel, and the differential braking of the trailer. By integrating the EHCSS model into the DAT's 3 DoF model, a new non-hierarchical model is established. The AOA is used to identify the parameters of this hierarchical model. The optimization results demonstrate that the new model is effective for developing active steering functions based on EHCSS. Additionally, torque vector control for the tractor and differential braking

control for the trailer are implemented as the proposed control strategy under jackknifing conditions. The control strategy is verified by HIL experiments under various working conditions. Experimental results indicate that the control strategy performs well in lateral stability control under different vehicle jackknifing conditions. Comparison with five other strategies reveals the feasibility and effectiveness of the proposed control strategy. Future work will involve conducting comprehensive vehicle tests to validate the control strategy in real scenarios. The insights gained from these practical experiments will be crucial for the practical deployment of the proposed control strategy in DATs.

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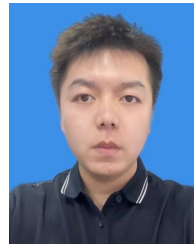
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